

Class Anthozoa

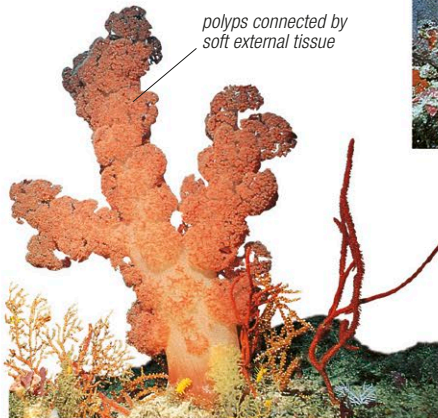
Anthozoans

Occurrence About 7,000 sp. worldwide; on the seabed, in soft sediments

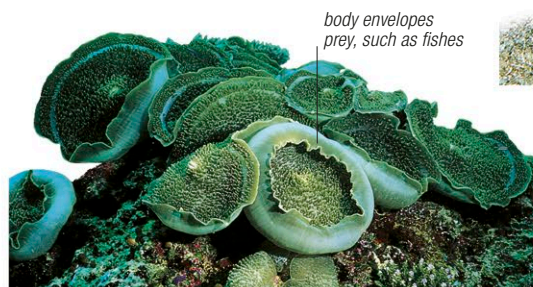


The word “anthozoa” means “flower animal,” and with their bright colors, waving tentacles, and fleshy trunks, these animals often look remarkably like plants. However, they are carnivorous—as are all cnidarians—although some species supplement their diet with food produced by microscopic algae harbored in their bodies. Anthozoans vary greatly in size, ranging from minute species less than $\frac{1}{32}$ in (1 mm) across, to those over 33 ft (10 m) across. When adult, they form polyps—simple, tubular animals with the tube closed at one end and open at the other. The open end is drawn out into one or more whorls of tentacles containing stinging cells, used for feeding and for defense against predators as varied as gastropods, polychaetes, sea spiders, and starfish. Anthozoans multiply asexually and sexually. Many species are solitary, but this group also includes colony-forming hard corals—animals that are

responsible for building coral reefs, the biggest structures in the living world. These are formed by the animals’ external calcareous skeleton. Other widespread anthozoans include soft-bodied sea anemones, plus sea pens and sea fans that have flexible, horny internal skeletons.

**BUSHLIKE SOFT CORALS**

The fleshy masses of most soft corals contain calcareous structures (sclerites) that support them. In the bushlike species (above), the sclerites may protrude when the colony is expanded. A member of the genus *Dendronephthya*, this Indo-Pacific coral grows to $3\frac{1}{4}$ ft (1 m) high. Soft corals like these are either attached to hard surfaces or anchored in soft sediments.

**FALSE CORALS**

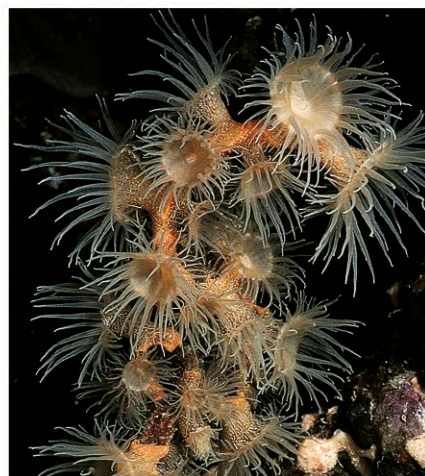
Living in tropical waters worldwide, false corals are intermediate in form between true corals and sea anemones. Their polyps closely resemble those of corals, but lack an external hard framework. Like reef corals, they have symbiotic algae that supply them with nutrients. *Amplexidiscus fenestrafer* (above) attains a diameter of at least 12 in (30 cm).

COMMON DEAD MAN'S FINGERS

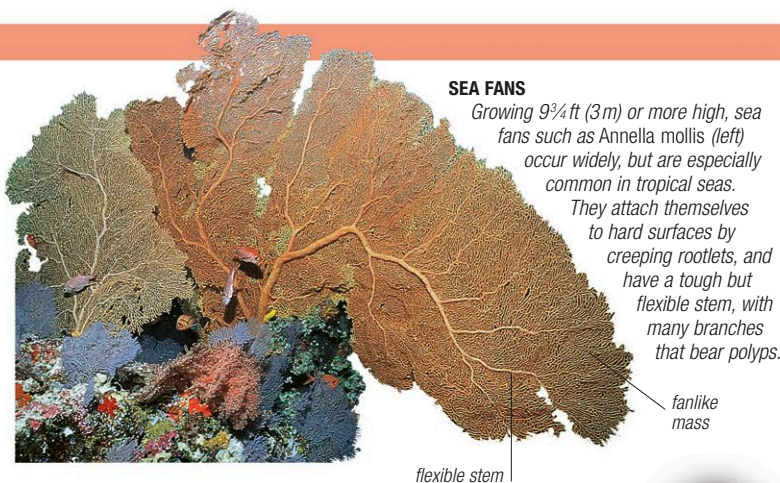
Soft corals—members of the order Alcyonacea—occur in all oceans. The North Atlantic species *Alcyonium digitatum* (above) grows up to 8 in (20 cm) high, and lives in shallow water. Its polyps protrude from a white to pink or orange, fingerlike mass—hence its ghoulish common name. The lobes contract if touched.

**REEF SOFT CORALS**

Like dead man's fingers, *Sarcophyton* species belong to the order Alcyonacea, and have the same fleshy texture. Members of this genus are most common in warm parts of the Indian and Pacific oceans, and form colonies up to $3\frac{1}{4}$ ft (1 m) across.

**HORMATHIID SEA ANEMONES**

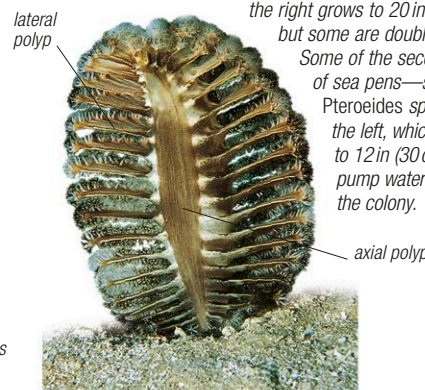
These widespread sea anemones commonly form symbiotic relationships with other marine invertebrates, living on mollusk shells or the carapaces and even claws of crabs. The anemone gains food scraps, while its partner is protected by the anemone's stinging tentacles. The species above, *Stylobates aenus*, grows up to 8 in (20 cm) tall.

**SEA FANS**

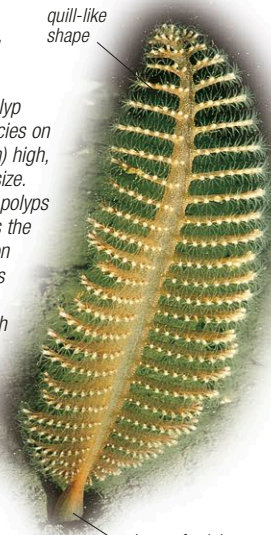
Growing $9\frac{1}{4}$ ft (3 m) or more high, sea fans such as *Annella mollis* (left) occur widely, but are especially common in tropical seas. They attach themselves to hard surfaces by creeping rootlets, and have a tough but flexible stem, with many branches that bear polyps.

SEA PENS

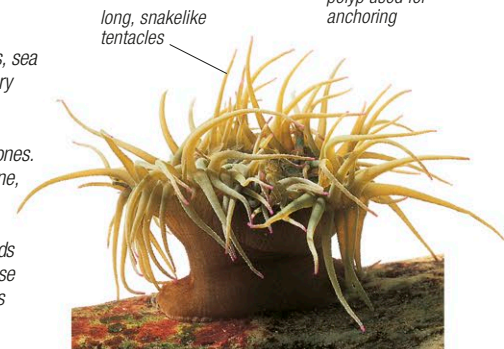
With a central axial polyp bearing many lateral polyps, often on regularly arranged side branches, sea pen colonies—found throughout the world—look like old-fashioned feather quills. The lower end of the axial polyp anchors the colony to the seabed. The *Pennatula* species on the right grows to 20 in (50 cm) high, but some are double this size.



Some of the secondary polyps of sea pens—such as the *Pteroeides* species on the left, which grows to 12 in (30 cm)—pump water through the colony.

**SEA ANEMONES**

Unlike most anthozoans, sea anemones live as solitary polyps; many of these, however, reproduce by dividing and forming clones. The snakelocks anemone, *Anemonia viridis* (right), has tentacles up to 4 in (10 cm) long. It needs well-lit conditions because a large part of its food is supplied by symbiotic photosynthetic algae.

**ANEMONIA ALICEMARTINAE**

Described as a new species in 2001, this bright red anemone is found on the rocky coasts—including tidal pools—of northern Chile. Scientists suggest that such a conspicuous species has not been spotted before because its population has only recently expanded over the last decades.

